Consistency Models

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Abstract

[PENDIENTE]

1 Introduction

Diffusion models [1, 2] are currently one of the most successful families of generative models for images. They gradually destroy the data by adding noise and train a neural network to reverse this process. The price of this quality is the cost of sampling: generating a single image requires starting from pure noise and removing it step by step, evaluating the network at every step. Ho et al. [1] use 1000 steps per sample. Formulating the process as an ordinary differential equation, the *probability flow ODE* [2], greatly reduces this number, but tens of network evaluations per image are still needed.

To address this problem, Song et al. [3] introduced *Consistency Models*. A consistency model learns to map any point on a trajectory of this ODE directly to its origin, that is, to the clean image at the end of the trajectory. As a result, it can generate a sample with a single network evaluation. When more compute is available, it can also chain several steps, adding noise back and removing it again, to improve quality. The paper proposes two ways to train them: *consistency distillation* (CD), which distills a pre-trained diffusion model that acts as a teacher, and *consistency training* (CT), which trains them from scratch, without a teacher.

In this project we reproduce this method at a small scale on MNIST [4] and measure what is gained and what is lost when the number of sampling steps is reduced. MNIST was chosen because it is light enough to train several complete models within the project deadline and without a GPU.

With respect to the initial proposal, the project has been extended in one direction: besides the diffusion teacher, we trained a second teacher based on Rectified Flow [5], and also distilled it into a consistency model. This allows us to study how the choice of teacher affects the resulting consistency model.

2 Objectives

The main objectives set in the project proposal are:

1. Train a diffusion model to serve as the teacher of the consistency model.
2. Distill the knowledge learned by the teacher into a faster consistency model.
3. Compare the consistency model with the diffusion model in terms of sampling speed and sample quality.
4. Analyze the results: the speed-up obtained, the quality of the samples generated with more or fewer steps and their diversity, as well as other technical issues of interest and the limitations found during the project.

Once these objectives were met, the proposal considered three extensions, in this order: training the model with consistency training and comparing it with the previous ones; adapting the architecture to color images with CIFAR-10 [6]; and developing a class-conditional consistency model able to generate a given digit.

All four main objectives have been met. [PENDIENTE: decidir qué extensiones se incluyen en la entrega.]

3 Related work and state of the art

Diffusion models. Diffusion models were introduced by Sohl-Dickstein et al. [7] and became competitive with DDPM [1] and score-based models [8]. Song et al. [2] unified both views as stochastic differential equations and showed that every diffusion process has an associated deterministic probability flow ODE with the same marginal distributions. Karras et al. [9] later reorganized their design space (noise schedule, network preconditioning and ODE solver); their choices are the basis of consistency models and of our implementation.

Fast sampling. Several lines of work reduce the number of network evaluations. Training-free methods use better solvers for the probability flow ODE, such as DDIM [10], the second-order Heun sampler of EDM [9] or DPM-Solver [11], which reaches good quality in around 10 steps. Distillation methods train a student to imitate several teacher steps at once: progressive distillation [12] halves the number of steps in each round, down to 4 steps, at the cost of several rounds of training. Finally, flow matching [13] and Rectified Flow [5] learn a velocity field along nearly straight paths between noise and data, which can be integrated with fewer steps; Rectified Flow can be straightened further with *reflow*.

Consistency models. Consistency models [3] learn a direct map from any point of an ODE trajectory to its origin. They allow one-step generation and, unlike progressive distillation, are trained in a single stage, either by distillation (CD) or from scratch (CT). Song and Dhariwal [14] substantially improved CT with a pseudo-Huber loss and a new discretization curriculum. Later works apply the idea to latent text-to-image models [15], learn jumps between any two times of the trajectory [16], and move to continuous time to scale consistency models to 1.5 B parameters [17]. All of them rely on large GPU budgets; our goal is instead a small reproduction on CPU, in which we also compare a diffusion model and a Rectified Flow as teachers.

4 Methodology

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4.1 Data and architecture

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4.2 Teacher models

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4.3 Consistency models

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4.4 Sampling from consistency models

[PENDIENTE]

4.5 Differences from the original paper

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4.6 Evaluation protocol

Reference classifier. Metrics based on Inception, such as FID, are designed for natural images, so we trained a small CNN on MNIST instead: two 3×3 convolutional blocks (32 and 64 channels, ReLU and 2×2 max-pooling), a hidden dense layer of 128 units and a 10-way output, with 421 642 parameters in total. It was trained for 10 epochs with Adam (learning rate 10^{-3}, batch size 128) and reaches 99.00% accuracy on the MNIST test set.

Quality and diversity metrics. Our main quality metric is FD-MNIST: the Fréchet distance between Gaussians fitted to the 128-dimensional penultimate-layer features of generated and real images. It is computed exactly like FID [18], with our classifier in place of Inception and 10 000 MNIST training images as the real reference. We also report the fraction of *recognizable* samples, those classified with a top-1 probability of at least 0.9, but only as a secondary metric: the classifier is over-confident outside the data distribution and assigns a digit with such confidence to 61% of pure-noise images. Diversity is measured by the entropy of the distribution of predicted digits, whose maximum, $\ln 10 \approx 2.30$, means that all digits are equally frequent. Since the Fréchet distance depends on the number of samples, every configuration (model and number of steps) is evaluated on the same $N_{\text{eval}} = 2000$ samples, generated from the same initial noise within each family (VE or RF). As references, we also evaluate 2000 real test images, which give the best achievable values, and 2000 pure-noise images.

Timing. We measure the wall-clock time needed to generate $N \in \{1, 10, 100, 500\}$ samples, averaged over 3 repetitions after a warm-up run, and report the time per image at $N = 500$. The speed-up of each configuration is computed with respect to the teacher of its family with the most steps. We also report the number of network function evaluations (NFE) per sample, which, unlike time, does not depend on the hardware. All measurements were made on the same CPU.

5 Results and discussion

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5.1 Quality versus number of steps

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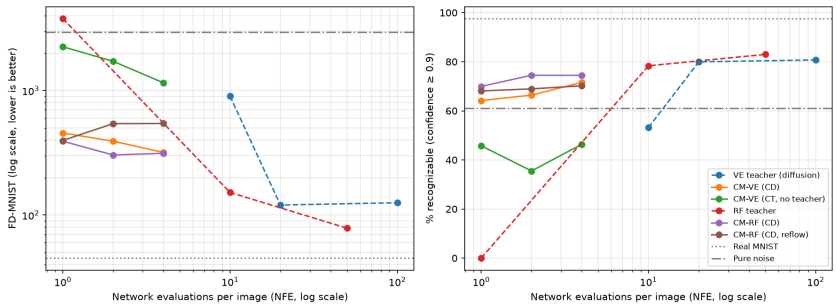


Figure 1: Sample quality versus number of network function evaluations (NFE): FD-MNIST (left, lower is better) and percentage of recognizable samples (right). Dashed lines are teachers and solid lines consistency models; gray lines mark the real-MNIST and pure-noise references.

5.2 Sampling speed

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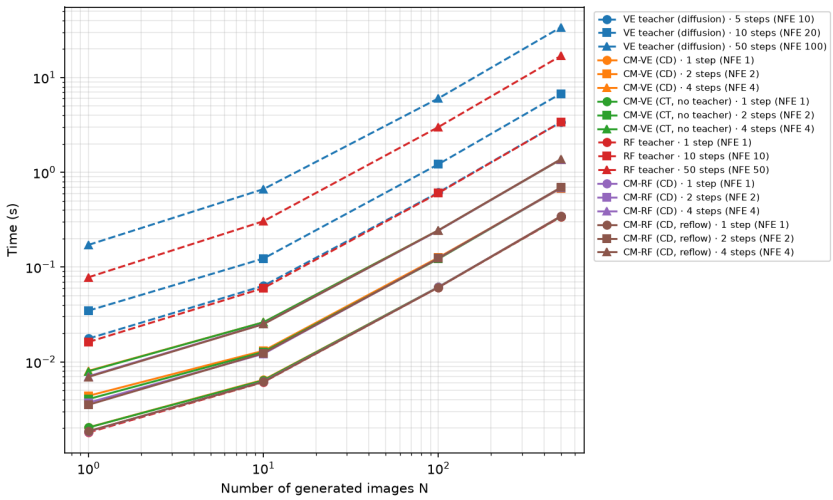


Figure 2: Sampling time versus number of generated samples N (log-log scale). Dashed lines are teachers and solid lines consistency models.

5.3 Diversity

[PENDIENTE]

5.4 Visual inspection

5.5 Discussion and limitations

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6 Conclusions and future work

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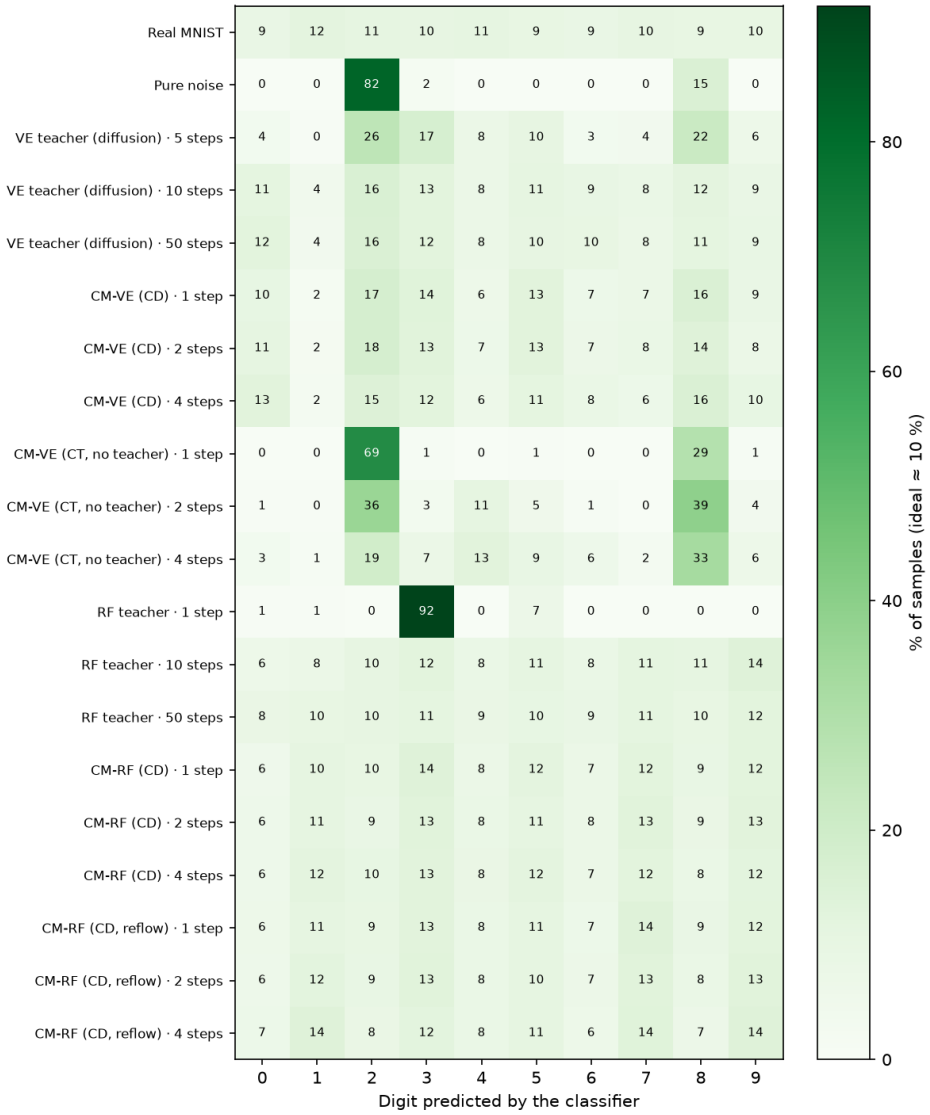


Figure 3: Distribution of the digits predicted by the reference classifier (percentage of samples per digit; a perfectly balanced generator gives 10% each).

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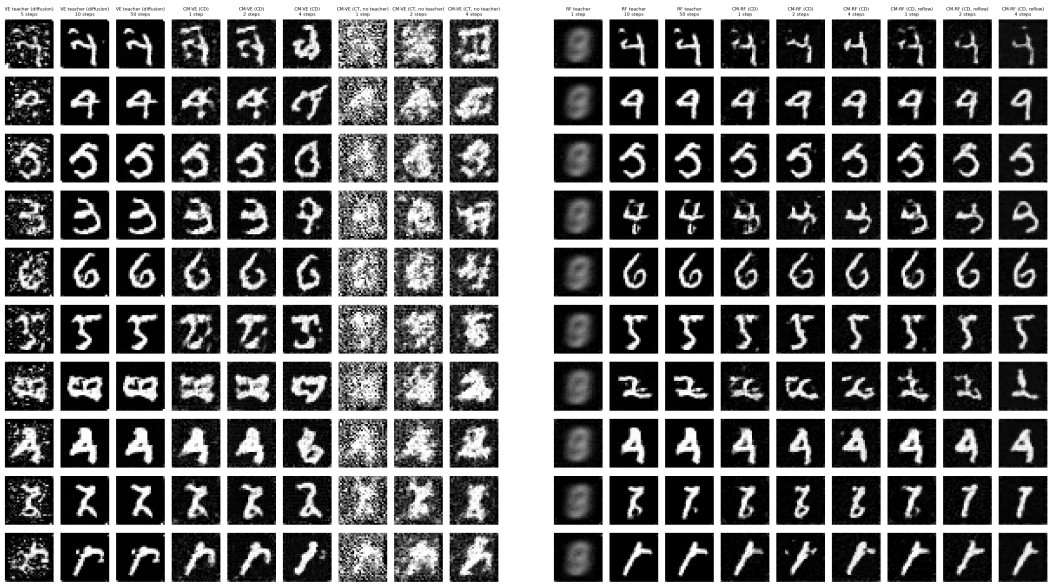


Figure 4: Samples from the VE family (left) and the RF family (right). Each column is a configuration and each row starts from the same initial noise.

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A Technical appendices

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Author contributions

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Declaration regarding the use of AI tools

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